

Review

A Literature Review on Energy Management Systems and Their Application on Harbour Activities

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Abstract

The growing global concern for sustainability and energy conservation has led to the adoption of energy management systems to minimise the impacts of energy intensive processes. This study reviews the evolution, the applications, and implementation techniques of energy management systems with an emphasis on harbour operations. Through the mapping of the research on energy management systems post-1973, the literature review demonstrated a substantial transformation of the systems from basic monitoring in the building sector to complex artificial intelligence analyses in smart and microgrids, industries, renewable energy sources integration, transportation, and harbours. Initial broad search (1973–2025) identified 22,003 EMS-related records; targeted port–EMS queries yielded 214 records, of which 139 unique records remained after de-duplication and 78 full texts were assessed. Finally, 27 studies were included in the quantitative synthesis. A meta-analysis in conjunction with an article review, and a weighted sum model coupled with sensitivity analyses revealed promising results for harbour energy management system implementation in terms of peak/load shifting, on-shore power supply, and real-time energy monitoring. The findings showed that energy management system efficacy is linked to maturity levels and strategic deployment of the measures/policies in each stage.

Keywords: energy management systems; harbours; sustainability; smart grids meta-analysis



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1. Introduction

During the last century, the profligate exploitation of fossil fuel resources to cope with the increasing demand of energy comprised the major factor of climatic change. The greenhouse phenomenon is enhanced by the large concentration of respective emissions in Earth's atmosphere, contributing to the increase in the average temperature of the planet [1,2]. Many global initiatives and agreements between states, such as the Kyoto protocol and the Paris agreement, aim to decrease the impacts of this environmental crisis through the adoption of sustainable forms of energy and smart management systems in contemporary energy applications [3,4].

Since 1990, greenhouse gas (GHG) emissions in EU-27 presented a decrease from 4922 Mt CO₂-eq. to 3484 Mt CO₂-eq. in 2022 [5]. Based on Figure 1, GHG emissions have been reduced in most sectors by the partial transition to renewable energy source (RES)-based technologies and the more efficient utilisation of the available resources. The energy sector, which in 1990 heavily relied on fossil fuel combustion, presented a significant reduction due to the integration of RESs, which mandated the use of smart management systems [6], while other sectors such as industrial processes, product use, and commercial services have

implemented additional measures consisting of energy monitoring and energy demand management [7].

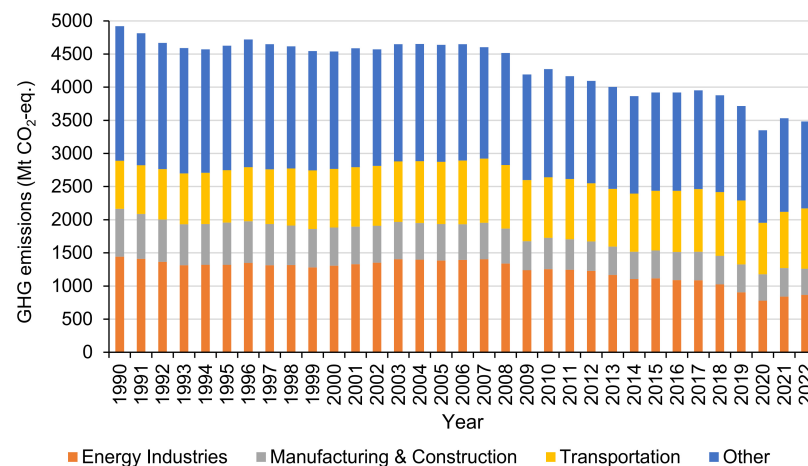


Figure 1. GHG emissions evolution in EU-27, 1990–2022, based on [5].

Since the early 1980s, when the electronic industry began to substantially develop computer-aided systems and applications, the energy demand planning known to be directly linked with energy management systems (EMSs) was studied as an energy control and management tool in many sectors, such as building, industry, and other energy-intensive applications [7]. Therefore, an EMS consists of a policy/technical framework where interacting elements aim to optimise the processes occurring in a specific field (i.e., building, microgrid, electrical network, etc.) in order to reduce operational costs and energy waste, improve stability and efficiency, and provide safety to application users [8–10].

Following the oil crisis in 1973, energy prices have significantly escalated, bringing to the surface significant concerns about energy preservation and cost minimisation. In the late 1970s, a great interest in developing EMSs to meet this target has been a goal for many corporations to secure the efficient use of energy and contribute to the financial status of end-user applications [11]. The first EMSs were initially documented in the scientific literature about 40–45 years ago and were related on the one hand to the management of building space conditioning and on the other to energy conservation and operational cost minimisation of industrial processes. Since 1990, apart from the above mentioned applications, EMSs are used in more complex cases, including RES power plants, electricity networks and microgrid operations, electric vehicles (EVs) powertrain, and small communities and ports' optimisation of processes [7].

Therefore, the evolution of EMSs could be divided according to Fawkes (2016) into five stages [12] (see Figure 2):

- From 1973 to 1981: Entitled as 'energy conservation', EMSs aimed at reducing the energy consumption in different applications by mostly altering, for example, the operating scheduling of processing in industry, or by setting some rules on the behaviour of occupants in the building sector. The advances in technology were introduced at a primary level, but this sometimes resulted in failures due to the early stage of development of the components.
- From 1981 to 1993: Entitled as 'energy management', this was the period when the first computer-aided monitoring and measurement systems were widely implemented. This early-stage-developed equipment facilitated the management of energy utilisation in applications accounting mostly of the building and industrial sectors through the integration of factors such as 'Degree Days' in space conditioning. Additionally, the

energy management market was established through the adoption of ‘Energy Service contracting’ and the occupation of ‘Energy Managers’.

- From 1993 to 2000: Entitled as ‘energy procurement’, this was mostly defined as the period when energy management was based on purchasing strategies, while supervisory control and data acquisition (SCADA) was introduced. EMSs based on advanced computers in electricity operators optimised energy transmission and distribution, allowing penetration of RESs in the electrical networks [13].
- From 2000 to 2010: This was entitled ‘carbon reduction’ due to the significant increase in RES capacity and the governmental approaches toward sustainability. EMSs evolved to be further applied to RES applications and microgrids in order to optimally allocate different energy sources without compromising reliability. In sectors where EMSs were implemented in the past (e.g., buildings), the initial objectives of energy conservation were adjusted to be able to integrate small-scale RES production and storage units aiming to support an environmentally friendlier action plan [13,14].
- From 2010 to the 2020s: This was entitled ‘energy efficiency’, with the main scope being the optimised operation of all energy-related processes with minimisation of the environmental impacts of fossil-fuel dominated sectors. EMSs were adopted by ‘green’ transportation, with their implementation in EVs, on the one hand to regulate the efficient utilisation of stored energy and optimise vehicle performance, and on the other to allow an interconnection to charging stations by controlling the energy flow in and from the EVs’ battery banks depending on the connectivity mode [15,16].

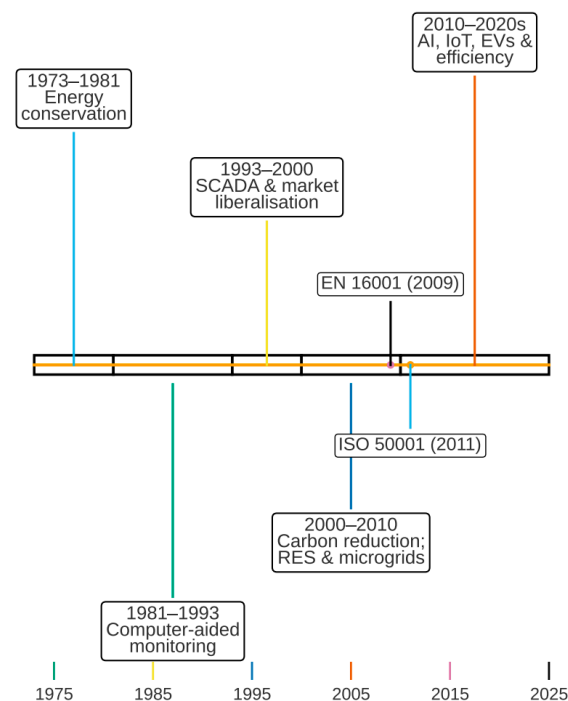


Figure 2. Five-stage EMS evolution. Based on [12].

The International Organisation for Standardisation (ISO) has issued the ISO 50001 [17] certificate in 2011, entitled ‘Energy management systems—Requirements with guidance for use’, as an overall guide to organisations to implement efficient energy performance actions as a continuous operational process instead of an individual project [18].

The ISO 50001 standard offers a framework for improving energy efficiency in different sectors, due to focusing away from technology-based actions on management practices to those of Backlund et al. (2012), and is a keystone to exploit the full potential towards a high

energy performance [19]. The guideline on which the 50001 standard is based comprises four main steps, described by a PDCA approach (Figure 3) [20]:

- Plan: Identify objectives, set targets, and make plans toward improving energy performance. This could be achieved via energy review and assessment of energy indicators.
- Do: Implementation of the plan—the participating entities should be trained and present awareness of the plan in order to control the operation.
- Check: Operation of the monitoring, measurement, and analysis system that allows an internal audit and provides corrective and preventive actions that improve energy performance.
- Act: Review of the energy management operations to re-assess the energy policy with a scope to further improve the energy performance of an organisation.

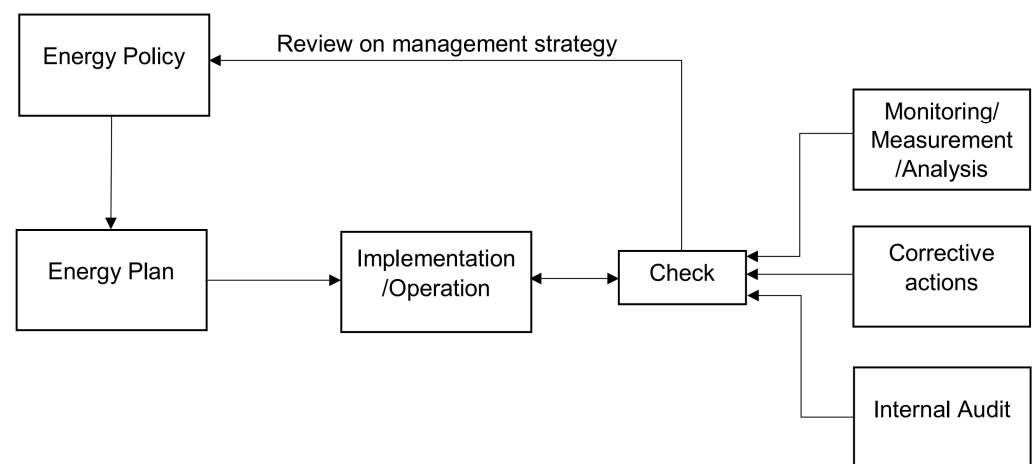


Figure 3. EMS implementation steps based on ISO 50001, based on [20].

The necessity of adopting energy conservation measures as discussed above to reduce on the one hand monetary losses and on the other promote sustainability has been imposed as a pathway to overcome the barriers of improving the energy footprint of different sectors. To this end, this research aimed at gathering the most relevant studies concerning the implementation of EMS strategy and integration in different sectors and applications to support the transition to a neutral carbon footprint. The objective of this study therefore is expressed through the below stated research questions (RQs):

- What are the applications and services of EMSs, and how are these implemented into business task scheduling?
- How could EMSs be implemented in harbour activities? What are the benefits and possible challenges?

To address the quoted RQs, the next sections of this study include the methodology that has been followed to gather the relevant data and to identify the applications and attributes of an EMS. Subsequently, the next sections present the main indicative results from the international peer reviewed literature that has been analysed to provide the answers to the stated RQs. Figure 4 shows the main steps of this review roadmap.

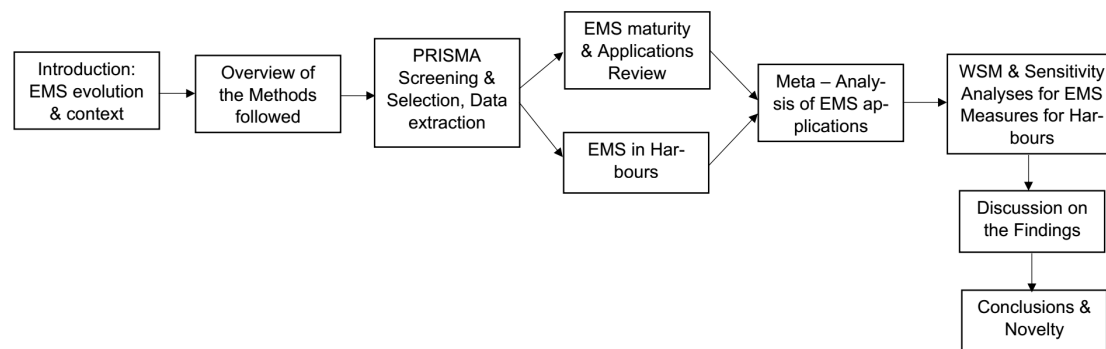


Figure 4. Structure of the steps followed during the literature review process.

2. Materials and Methods

Examining the above-stated RQs was based on an approach comprising three basic steps [21]. Initially, the exploratory phase was conducted, where the interpretation and identification of the role of EMSs provided the criteria used to locate the relevant information in peer reviewed literature, organisations, and technical companies. Secondly, the review process was conducted based on finding the available data on EMS technologies research to this day by using a keywords including ‘Energy Management System applications’, ‘EMS applications’, ‘Building Energy Management’, and ‘Smart Ports’.

During this stage, thorough research was carried out to ensure that the most relevant articles were retrieved. Exclusion criteria were imported in the methodology, which consisted of studies where EMSs were a minor component, non-peer-reviewed sources such as blogs and websites, and articles where technical details were absent. Furthermore, according to the above-mentioned data on EMSs evolution, the research was limited to resources following the 1973 oil crisis, while focusing more on documented articles and reports since the 2000s due to the high number of published articles.

Due to the high number of studies including EMS applications, there was an extensive overview of the results to obtain an indicative number by decade (i.e., 1980s, 1990s, 2000s, 2010s, 2020s) that investigated different applications and services.

The final step of the followed methodology, used to support the second RQ, incorporated inclusion criteria on the results yielded from the previous stage. Those criteria took into account studies with EMSs applied specifically to maritime ports and shore-to-vessel power, along with other applications relevant to harbour activities, such as transportation services.

To categorise the content of each section of this study, the assessed data yielded from the comprehensive literature review have been documented in parts concerning the EMS applications and their benefits; their implementation maturity, indicating the levels of their integration; and, finally, their application in harbour activities. Therefore, this study employed a qualitative–quantitative approach, integrating document analysis with meta-analytical techniques.

Figure 5 depicts the phases of the review process dictating how this research has been carried out.

The collection resulted in 334 articles, book chapters, and conference abstracts within the field of engineering between 1973 and 1980, from which only few were analysing the applications of EMS, and 900 between 1981 and 1990, which included reviews on EMSs; however, likewise to the previous decade, most of those were dedicated to computer engineering and not their implementation in energy conservation. Between 1991 and 2000 (engineering and energy), 1538 studies have been identified within engineering and energy applications, 1392 between 2001 and 2010, and 17,839 between 2011 and 2025. By

incorporating EMSs for harbour applications, the number of peer-reviewed literature was found to equal 149 since 2010, with most of them comprising offshore vessel applications and therefore indicating that it is a topic that has been recently investigated in the scientific community.

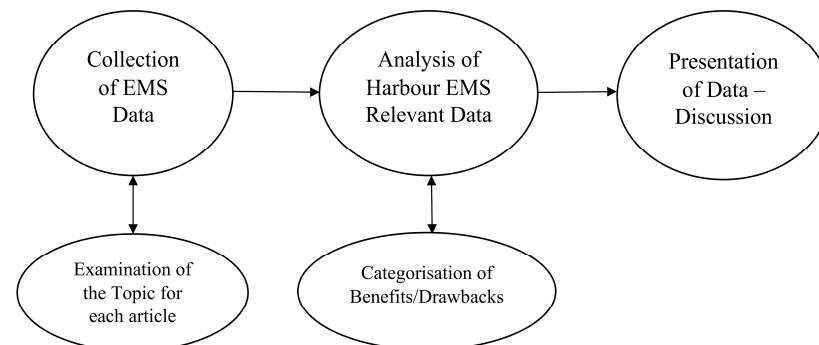


Figure 5. Methodology of the review process, based on [22].

Furthermore, the review followed PRISMA-2020 [23] to enhance transparency and reproducibility of the research, screening, and eligibility steps that precede the quantitative synthesis. To align with the decade mapping already reported, an initial broad, uncombined search across 1973–2025 yielded $n1 = 22,003$ records (general corpus).

For the evidence synthesis, targeted queries were then applied in ‘Scopus’, ‘IEEE Xplore’, and ‘OpenAlex’ databases, using three validated keyword clusters: (i) port context (port/harbor/harbour, ‘container terminal(s)’); (ii) EMS terms (‘energy management system(s)’, EnMS, EMS, ‘ISO 50001’, ‘energy management’); and (iii) EMS-operable measures (‘shore power’/OPS/‘cold ironing’, electrification, microgrid(s), demand response, peak shaving, load shifting, ‘smart port(s)’, ‘digital twin(s)’). The results provided $N1 = 214$ articles written in English within the field of engineering and energy. The clusters were piloted against studies already cited in the manuscript; those seed studies and their citation chains were included at identification prior to de-duplication. After cross-database de-duplication, $N2 = 139$ unique records remained and their titles/abstracts were screened, with 61 exclusions (primarily out of scope, 46; no EMS/EMS-operable measure, 15). $N3 = 78$ full texts were assessed, and $N4 = 27$ studies were included in the quantitative synthesis. The PRISMA 2020 statement and results are depicted in Figure 6.

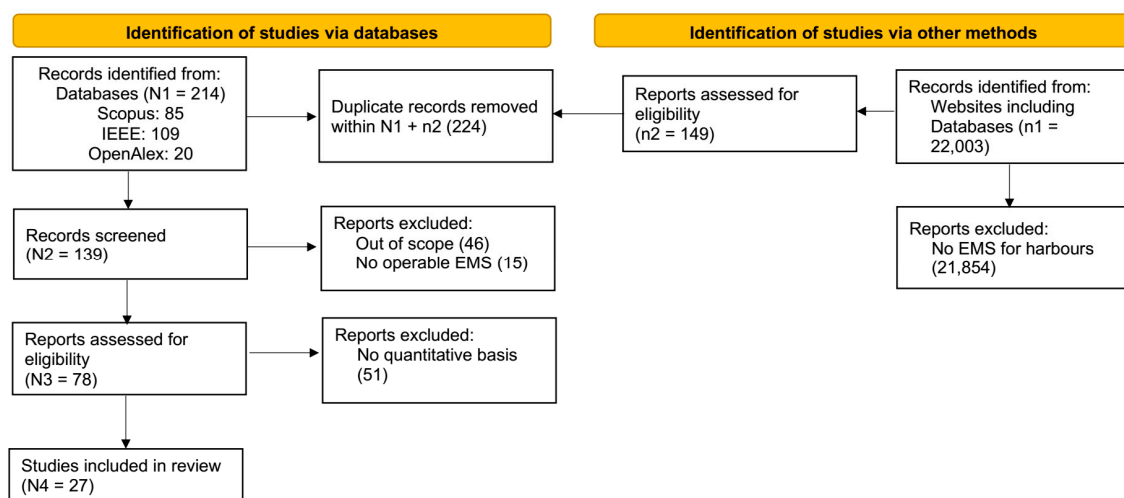


Figure 6. Review process for EMS in harbours based on the PRISMA 2020 statement.

Since the identification of the articles to be included in this review, a data extraction checklist, presented in Table 1, was used to gather the necessary information to perform the subsequent analysis.

Table 1. Data extraction checklist of the articles reviewed.

Category	Fields to Extract
Bibliographic	Authors, year, venue, DOI
Context	Sector/site (port/terminal/utility/building), geography, time window
EMS type and measures	Monitoring/SCADA, OPS, RES, storage, electrification, demand response
Outcomes	Energy (MWh, %), cost (CAPEX, payback, LCOE), emissions (tCO ₂ , intensity)
Baseline/comparator	Normalisation (per operation/throughput), counterfactual definition
Methods/statistics	Study design, data sources, methods followed
Standards/policy link	ISO 50001, EN 16001, PDCA cycle

To quantify the evolution of the EMS publications, a decade-wise diagram was plotted showing the growth of the EMS-based literature through the last 50 years (see Figure 7). In this context, the performance of EMS implementation was assessed through a meta-summary of the data presented in the next section in order to enhance the empirical relevance of EMS applications. In this context, to appraise how EMS implementation affects the energy consumption in harbour applications, a meta-analysis consisting of the article corpus given by PRISMA 2020 was performed. In the case where the results of the articles presented percentages in energy conservation, (e.g., “−p% energy”), effects were expressed on a multiplicative scale using the log response ratio, $\ln RR = \ln(1 - p/100)$, while, when there were absolute numbers of energy conservation without a baseline, this was only listed [24]. Additionally, a cost-effectiveness analysis of the studies included in the review (N4) (where applicable) by PRISMA 2020 was performed. For each harbour case, costs and savings were normalised to simple and comparable indicators, such as simple payback (CAPEX/annual savings) and—where emissions were reported—an implied annual benefit per tonne of CO₂ (annual cost savings/annual tCO₂ avoided).

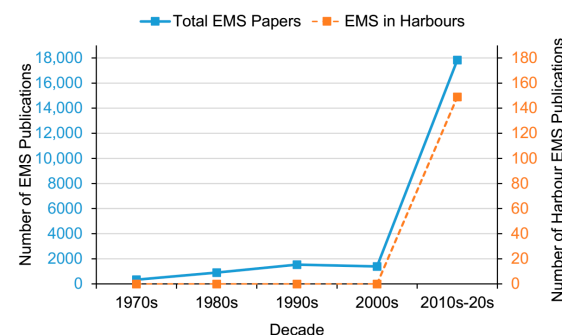


Figure 7. EMS publications' evolution.

Finally, to evaluate the technologies implemented in EMS harbour applications, a weighted sum model (WSM) [25], which is also known as multi-criteria analysis, has been applied according to the Equation (1).

$$a = \sum_{i=1}^N W_i X_i \quad (1)$$

where a represents the score of the benefits and challenges, N is the number of the evaluation criteria (C_i), W_i is the weight of the criteria C_i , and X_i is the value function of criteria C_i .

To assess the robustness of multi-criteria results, a sensitivity analysis was conducted on the inputs and modelling choices. The analysis varied only elements that are plausibly uncertain or preference-driven and then observed how rankings changed. Those elements included on the one hand the weights of criteria and on the other the value function (score) of each criterion.

3. Energy Management Systems Applications

As discussed in the previous section, the adoption of EMSs by parties that either are characterised by energy intensive processes or are involved in the energy market could be deemed of great importance towards the transition to sustainability. Although EMSs have been established as an official policy in several countries and an international standard since the 2000s and 2010s, respectively [26], their history goes back many decades to after the 1970s energy crisis, when governments and organisations realised the importance of energy conservation and economy.

Berlad et al. investigated the implementation of thermal storage combined with an efficient operational strategy of the space conditioning processes of a residence located in New York, USA. The results of their research showed that the energy management scheme allowed a 50% decrease in the operating costs without compromising the life-style of the occupants [27]. An approach to reduce the electricity costs of a residence based on a computer–energy control system, taking into account the time-dependent electricity costs, was studied in 1982 by Capehart et al. The authors documented cost reductions of USD 140 by using an optimisation algorithm to manage the energy demand of the residence’s appliances and the occupants’ behaviour based on the set electricity prices by the utilities [28].

The first steps of EMS implementation in the UK’s industry sector were quoted by Lewis, who identified the potential of energy savings in industrial processes. The energy conservation measures proposed by the author included targeted modifications in several industrial sectors by altering either the processes or by implementing a management of the existing ones to secure less consumed energy with lower anticipated expenditures [29]. In the same context, Kaya and Keyes presented in their study the structure of an EMS, along with its respective implementation levels based on the processes of pulp and paper industries. The authors identified the steps and measurements that, through the implementation of micro-processors in a plant’s production facilities, would optimise the operation of the systems and reduce energy consumption [30]. An initial approach of EMS implementation in electrical utilities was investigated by Gellings, who discussed the planning of demand-side management techniques to shape the utility’s load and provide energy conservation. The author emphasized end-user behaviour and efficiency integration [31].

The use of microcomputers in EMSs was adopted substantially at the end of the 1980s and early 1990s due to the advances in chip processors and cost reduction of the electronic components. Benator described the benefits of using early-stage computer-aided programs in EMSs to log, calculate, and graphically present energy consumption patterns. Additionally, these systems could identify cost-efficient energy conservation opportunities and set milestones for advanced energy management [32]. A similar approach to calculate

and evaluate the thermal energy profile of buildings or urban district heating systems was documented by Taesler, who incorporated a software package entitled 'ENLOSS' in order to forecast the energy demand in the short term and optimise indoor space conditioning [33]. EMSs were utilised in electricity networks to avoid voltage disturbances and overloads. An EMS combined with an expert system was studied by Handschin and Hoffmann in a 110 kV utility's electrical network. The results indicated that the system was able to automatically locate all failures to affect the network's stability and therefore avoid overloads and voltage limit violations [34]. The benefits of implementing EMSs in electrical utility companies to provide monitoring of energy consumption were assessed by St. Clair. The author emphasized the importance of monitoring and targeting techniques through an EMS in many UK sectors, including large- and medium-sized industries and small and medium commercial enterprises (SMEs). It was quoted that, in this context, 25–35% of energy savings could be achieved on an annual basis [35]. Gruber and Brand pointed out the importance of energy management and potential savings in SMEs in Germany, showing conservation between 40% and 60% depending on the measures adopted and the field of operations. However, as the size of the company was smaller, the level of EMSs implemented was lower, providing less savings in both fuel and electricity consumptions [36].

Since the 2000s, the electricity markets of many countries were liberalised, resulting in a significant increase in the energy data distributed to the relevant participants by the transmission and distribution system operators (TSOs, DSOs) [37]. Werner et al. presented in their research an EMS framework able to balance the load profile of consumers via a code-programming model in order to secure a quick and secure relocation of data to all market participants in the German electricity market [38]. Moreover, according to Celli et al., the continuously increasing integration of RESs in electricity networks and microgrids is a major obstacle for transmission and distribution system operators. The authors investigated the application of a neural network (NN)-based EMS into a microgrid to optimise the dispatchable units operation to meet the load demand with cost, losses, and emissions minimisation [39]. El-Shatter et al. presented a study of optimising the operation of an energy system consisting of RESs and hydrogen electrolysis—re-powering topology via the implementation of a Fuzzy logic-controlled EMS. The simulated operation of the configuration aimed at regulating the power flow between the microgrid components to satisfy all load requirements [40]. Likewise, Kyriakarakos et al. published their study on a Fuzzy logic-based EMS for a simulated microgrid, including RESs, energy storage configurations, and a desalination unit. The results showed that the EMS optimised the energy flows and allowed decreased component sizes to meet the corresponding demand [41]. To cope with the intermittency and the dynamic operation of RES, Zia et al. analysed and compared multiple decision-making strategies for microgrid EMSs. The main purpose of the specific EMSs was quoted by the authors to include a multi-dimensional optimisation of microgrid operation in regard to increased efficiency and reliability, and decreased costs, energy losses, energy consumption, and GHG emissions. The strategies identified by the authors were divided into two stages comprising all associated costs (e.g., levelised cost of energy, operational and maintenance costs, etc.), and limitations of the system (e.g., network constraints, energy balancing, physical limits, etc.) [6]. An investigation of a hybrid EMS for microgrids consisting of RESs, fuel cells, and electrolyzers was conducted by Romero et al. The results indicated on the one hand a decreased energy loss and consequently a 28% efficiency improvement, and on the other an increased grid stability via minimising system stress by dispatching power accordingly [42].

The new status of energy systems in either micro- or nano-grids consisting of RES-based generation, energy storage configurations, and EVs mandates the adoption of advanced EMSs to facilitate the efficient operation of the respective components. Leonori et al.

suggested a new approach in regard to the operation of an EMS in real-time scheduling of energy flows in micro- and nano-grids. The EMS was based on a Neuro Fuzzy interface presenting negligible computational requirements compared to a Rolling Time Horizon EMS, which performed well only when the supporting prediction systems provided adequate inputs [43]. Under a similar scope, Roslan et al. investigated an optimised EMS controller for managing the varying load conditions of Perlis, Malaysia, to minimise operating costs, reduce GHG emissions, and cope with the complicated optimisation constraints. The results indicated that the introduction of the controller proposed by the authors decreased the costs and the emissions by 62.50% and 61.98%, respectively [44]. Ren et al. have documented a novel approach for EMS support by using Internet of Things (IoT) technology. The authors argued that IoT, combined with current computer software and programming languages, is able to succeed in more efficient performances of EMSs, with response time and CPU utilisation less than 5 s and 70%, respectively, due to the advanced communication protocols used [45]. Shehzad et al. proposed an EMS to ensure the most efficient CO₂ capture system operation coupled to an offshore wind farm and a battery energy storage topology. The EMS was developed accordingly to efficiently manage the stochastic production of the wind farm and the direct air capture loads [46].

The integration of IoT in EMSs is cited in the international literature to be of major importance in the buildings sector too. Bellagente et al. investigated the opportunities of integrating IoT in an EMS of the University of Brescia campus. The EMS interface consisted of several sensors, including energy meters, temperature sensors, and a primary substation logic connected to the DSO's SCADA. The IoT approach facilitated the data provision among several end users accounting of humans and digital artefacts [47]. The interactivity of EMSs in smart buildings was studied by Marinakis and Doukas. The IoT-based EMS system proposed by the authors combined a series of components, including data collectors in regards to building energy consumption, weather data, energy prices and end users' behaviour. The data processing predicted the energy behaviour and suggested actions to improve the energy performance of the building. The case study revealed significant energy savings, allowing a payback period of the investment of approximately two years [48]. Under the scope of smart building energy management, Zhang et al. used a mixed integer linear programming approach to minimise cost and energy demand. The scenarios they developed included different operating habits of end users living in a smart building, with multiple residences connected to the grid and to wind and combined heat and power generators operating under different conditions. The results documented cost savings between 11 and 23%, and peak demand savings of 5–59% depending on the case [49]. A simulation tool for the effectiveness of EMSs in the building sector was presented by Karki et al. The developed model provides results on the investigated building's energy performance, based on inputs including the EMS controls and the building's operational characteristics such as energy demand on space cooling and heating [50]. Korus and Jabłoński published a study integrating a modular IoT-based EMS within industrial facilities. The authors presented a case study where real-time data acquisition from a university campus environment was used as input to the EMS in order to improve response time to energy anomalies [51].

In the context of hybrid energy systems such as the one studied by Navarro et al. consisting of a proton exchange membrane fuel cell (PEMFC)-based micro combined heat and power (CHP) residential system, EMSs have shown positive results in terms of operating and financial potential. The EMS investigated by the authors has shown high adaptability in different operating scenarios, while at the same time reducing the investment's payback period and overall cost-effectiveness [52].

The application of EMSs to reduce energy consumption and promote a 'green' environmental footprint in ports has been studied during the last few years. Boile et al.

documented that minimising energy consumption and consequently improving the carbon profile of ports is a major concern among European port stakeholders, which has led to the adoption of measures including the implementation of EMSs. Koper port installed a EUR 65,000 EMS in one terminal resulting in 250 MWh annual energy savings accounting for EUR 19,500, and a reduction of 125 tCO₂ annual emissions [53]. Additionally, Iris and Lam reviewed the possible operational strategies and EMS opportunities for providing increased efficiency in ports' processes, low costs, and reduced energy consumption. They documented measures that would enhance, among other things, the potential of ports towards sustainability [54]. In the same context, Iris and Lam stated that many ports aim to adopt EMSs, due to high energy prices, that would enable the electrification of their processes and incorporate the use of RESs coupled with energy storage systems instead of carbon-intensive solutions. A port's microgrid, therefore, would require a smart EMS able to facilitate the energy flows between energy sources (i.e., local production and utility grid), while at the same time optimising the energy load support and equipment's efficient operation. The results indicated that costs could be significantly minimised with the adoption of the EMS plan [55].

During the last decade, the development of the EV market expanded significantly, with most automobile manufacturers investing in either plug-in hybrid EVs or battery EVs. EMSs are a crucial component in the EV drivetrain, as it could provide an optimised flow of the stored electricity and support battery storage reliability. Hannan et al. reviewed the different types of Li-ion batteries for EVs and documented the services provided by the battery EMS of an EV. The EMS controls the charging and discharging processes, while at the same time monitoring the operating temperature, the voltage and current input/output, and the battery cell's condition. Additionally, it provides data acquisition and assesses potential faults [56]. A Fuzzy logic control EMS based on the driving pattern recognition of an EV with battery and supercapacitor energy storage was assessed by Hu et al. The use of the Fuzzy logic-based EMS provided a battery lifetime extension by approximately 6.2% due to the decrease in the maximum charging/discharging current by 58.2%, while the vehicle's range increased by approximately 11.1% [57]. Guo et al. presented the benefits of a real-time NN model predictive control-based EMS for plug-in EVs. The real-time energy management method was able to calculate the battery's state-of-charge trajectory planning based on the near-future driving cycle predicted by the NN, and accordingly regulate the energy flow in the vehicle's drivetrain. In this context, the outcome showed a reduced fuel consumption by 5.1% compared to the rule-based control strategy [58].

Huang et al. reviewed and analysed contemporary AI-driven EMS strategies for hybrid EVs, identifying that model predictive control and reinforcement learning emerge as dominant technologies. These core technologies have been found to improve fuel efficiency and emission control [59]. As is apparent, the implementation of 'smart' artificial intelligence (AI)-based EMSs in contemporary applications, such as EVs or others considered as microgrids, could be deemed as a major step towards sustainability. Hamidi et al. documented a review on alternative AI-based EMSs. Four main categories were identified with their advantages and disadvantages; expert systems, although they present program flexibility and reliability, and operate with incomplete data, have a high development cost. Machine learning, in contrast, has lower cost and is reliable and flexible, but has a requirement of many labelled data. On the other hand, artificial NNs present several benefits, including self-learning, simplicity of implementation, flexibility, and no requirement of mathematical models, while also having drawbacks such as training requirements and a processing time that, depending on the case, might be long. Lastly, Fuzzy logic-based EMSs are similar to human reasoning; no mathematical modelling is required and it presents

rapid operation. Nevertheless, it is not considered entirely stable, and it has a limited number of input variables [60].

Based on the above, the implementation of EMSs is one of the main factors for the operation of energy systems. Table 2 summarises the sectors and the respective end-user primary services that an EMS could be applied to and asked to provide, respectively.

Table 2. Applications and services of EMSs.

Sector	Energy Conservation	Cost Minimisation	Energy Flow Optimisation	Efficient Operation of Equipment	GHG Emissions Reduction
Buildings	✓	✓		✓	✓
Municipalities	✓	✓			✓
Industries	✓	✓		✓	✓
SMEs	✓	✓		✓	✓
Electricity utilities		✓	✓	✓	
TSOs/DSOs		✓	✓	✓	✓
RES installations			✓	✓	
Electric vehicles	✓		✓	✓	✓
Ports	✓	✓	✓	✓	✓

4. EMS Maturity Levels Framework

To assess the current situation of an organisation in regards to the implementation level of energy management, the introduction of maturity models has been identified as a tool for continuous improvement [61]. According to Kohlegger et al., ‘a maturity model represents phases of increasing quantitative or qualitative capability changes of a maturing element in order to assess its advances with respect to defined focus areas’ [62]. To this end, a maturity model could be deemed as a crucial method for the improvement of processes in several applications, and usually describes the maturity levels and the respective improvement measures through a five-point Likert scale, where five represents the highest maturity level [63]. Maturity model implementation has been widely adopted since the presentation of the Capability Maturity Model (CMM) by Paulk et al. to improve software processing [64], and is continuously developing in most key areas, e.g., Capability Maturity Model Integration (CMMI) in the fields of product and service development, services, and product/service acquisition [65].

The application of maturity models in EMSs is able to improve the understanding of energy management strategies, provide a roadmap towards continuous improvement, recognise the steps to be taken for successful implementation, and assess the maturity level allowing for a baselining [61,66]. Most contemporary EMS maturity models that are suggested as an industrial guideline or studied in academia are based on the PDCA framework (see Section 1 and Figure 2), as described in ISO 50001 [65]. The United Nations Industrial Development Organisation (UNIDO) issued, in 2015, a guide to EMS implementation for companies and industries analysing the steps to be followed for a successful integration of energy management into energy-related processes.

According to UNIDO, there are four main steps in compliance with the PDCA approach. First is preparation and commitment, which includes the identification of the existing level of energy management in order to set priorities and energy performance indicators (EnPIs), the development of the company’s energy policy, and the establishment of the main scope of the project along with the management team to implement EMS. Second

is energy management planning, where all legal and other types of requirements applicable to the organisation's use of energy are quantified, and energy review of past energy data provides patterns of energy use and consumption. Third is the operation of the system, which is divided in two sub-categories where on the one hand the personnel training and administrative routine (e.g., documents archived and communicated to relative parties) are secured, and on the other the efficient operation of the entire equipment is ensured. Last is the reviewing process, where monitoring of the energy metrics, along with scheduled audits, allow the re-adjustment of the EnPIs and the objectives set in the first step [67].

Ngai et al. proposed an energy and utility management maturity model to assess the level of implementation of EMSs in organisations and provide a guide towards advancement of EMS operation. The maturity model was based on CMMI, and involved five maturity levels and four transition phases between each maturity level. The first level implies that organisations have not implemented any EMS and no specific energy plan is under consideration. The second level entitled 'Managed' corresponds to the case where organisations manage the monitoring, control, measurements, and utilisation processes. The establishment of the energy management processes defines the third level across the organisation, while the practices and procedure are standardised toward sustainability. The fourth level implies an efficient energy and utility management, and the identification of process variation is achieved through the analyses of natural resource management collected data. The final level is described by the continual evaluation and improvement of the causes of process variations observed in the fourth level [68].

Antunes et al. presented their proposal on an EMS maturity level model based on the PDCA framework of the ISO 50001 as guidance to efficiently implement energy management and comply with the ISO certificates. Likewise, the authors described a five-scale maturity EMS model where, in the first level, no defined activities concerning energy management are documented. The second level corresponds to organisations' activities to assess the current performance via energy reviewing, identify EnPIs, establish an energy policy plan, and set objectives and targets. The implementation stage comprises the third level, where the organisation invests in equipment, training of the personnel, and documentation. The fourth level accounts for the monitoring stage where energy flows are monitored, and the received data is analysed, while energy audits are programmed to provide improvements in the initial planning through the fifth and last maturity level [61].

Similar to the aforementioned studies, Introna et al. suggested a maturity level model for energy management consistent with ISO 50001. Five levels of maturity were identified with five dimensions, cross-sectional to the documented levels, including awareness, methodological approach, energy performance management, organisational structure, and strategy. The initial stage is the level where the organisation is not managing energy flows; the second, entitled 'occasional', corresponds to the attempts of a minority within the organisation to collect information on energy consumption patterns and the identification of measures to occasionally reduce this consumption. During the third level of the maturity model, the organisation develops an energy strategy and sets specific targets and objectives aiming to be achieved through specific projects. EnPIs are defined and monitoring of consumption occurs in order to identify and bridge gaps in technical and managerial knowledge. The fourth level is described by the maturity of the company to comprehend that energy conservation is a continuous effort instead of a project-based venture. Monitoring the consumption is a standard activity, and the management team assesses the EMS through internal audits. In the last maturity level, the implemented EMS is optimally operating as a result of a continuous improvement and objectives' achievement that are subsequently disseminated outside the organisation, helping to reinforce its image [69].

In the same context, Jovanović and Filipović used an ISO 50001- and CMM-based maturity model, which has been applied to both ISO-certified and non-certified companies as a guide toward higher maturity levels. The proposed model included five maturity levels: 'initial', 'managed', 'defined', 'quantitatively managed', and 'optimised'. The 'initial' stage represents the level where the organisation has not implemented any energy management plans. The next level fits organisations where mechanisms for monitoring are implemented, some targets and plans are identified for specific processes, and ISO 50001 is partly adopted. At the third level, energy management is established, the processes are documented, and all ISO standards are implemented. The fourth level is characterised by a thorough analysis of the collected data and an expansion of monitoring to environmental factors too. To this end, some actions regarding environmental management systems (i.e., ISO 14001) are additionally integrated. The fifth level corresponds to the optimised operation of the EMS where all targets are met, and the energy-saving and environmental practices are continuously updated to cope with possible challenges [70].

Compared to the above presented research, Finnerty et al. published their study on an EMS maturity model focusing on multi-site or global organisations instead of site-focused cases. The maturity model proposed by the authors evaluates the performance of each organisation's site and the overall 'network' of sites to identify the technical and non-technical gaps to implement energy efficiency actions. The EMS maturity levels are listed as a five-point score, with the first one corresponding to no energy policy within the organisation and the second one defined by developing an energy policy plan based on an initial assessment of the organisation's energy performance. At the third level, the organisation has implemented the energy strategy and has begun taking measures to reduce energy consumption. The fourth level is described by a more consistent implementation of measures identified to improve energy efficiency and collaboration with local/national authorities, facilitating the set objectives. The last level of EMS maturity is the stage where the organisation is running a fully optimised energy management system consisting of a reference point for energy efficiency good practices [65].

Prashar developed and applied an EMS maturity model based on the ISO 50001 to SMEs, as the models existing in international literature were not applicable to small companies due to lack of resources and professional knowledge. The maturity model, similarly to the other ISO-based ones, categorised the levels into five scales with the first being defined by no energy policy and plans in the company. The second level incorporates an initial planning and monitoring/logging consumption concerning specific energy-intensive processes within the company, while the company's technical personnel is offered limited training. At the third level, a team within the company handles energy management issues, while energy efficiency matters with a feasible business case only are tackled. The fourth level implies the assignment of the energy management to a dedicated energy manager who supervises the planning and monitoring, and arranges the necessary investments to improve energy efficiency. Above all, the review of the overall processes to identify the entire potential of energy conservation is promoted. The last level is described by the optimum operation of the EMS, where energy efficiency consists of a strategic driver to identify market opportunities, and new innovative technologies are developed in both the company and the entire supply chain [71].

Acknowledging the above, it can be derived that, in most of the studies, two aspects of maturity levels are identified based on the following:

- The awareness, training, and task assignment of the organisation's team to secure an efficient EMS implementation and operation.
- The technical and the data processing implementation policy allowing an optimum assessment.

A summarised approach of the EMS maturity levels is depicted in Figure 8, accounting for a five-stage assessing scale where the lower level represents organisations with no energy conservation policies under consideration, and the top level describes organisations with the optimum operation of an EMS and a fully developed strategic plan that allows continuous assessment of the internal processes. The synopsis of the five identified EMS levels is presented in Table 3.

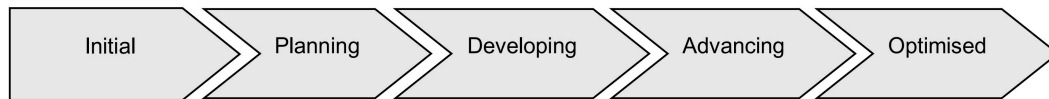


Figure 8. Maturity levels, based on [61,65,68–71].

Table 3. EMS implementation maturity levels and associated challenges, based on [61,65,68–71].

Maturity Level	Human Resources	Technical Implementation
1. Initial	None of the personnel is assigned to find solutions toward energy conservation.	No energy policy, and no technical background is available.
2. Planning	Targeted internal seminars to technical staff aim at teaching the basics of energy management.	An energy strategy is compiled with initial assessment of energy intensive processes to identify EnPIs and improvement opportunities.
3. Developing	A team is assigned to handle the EMS implementation, while participation in targeted projects aims to locate improvements in energy efficiency.	Following the energy plan, targets have been set and investments on the EMS and the respective infrastructure cover most processes.
4. Advancing	Responsibilities are identified at different levels in hierarchy—a project manager is appointed.	A fully monitoring/logging system is operational/collecting data from overall processes. Data processing, analysis, and energy audits provide insights for improvements.
5. Optimised	The entire personnel is involved in the EMS stages, results are disseminated to similar entities and the supply chain.	Energy management provides optimum energy conservation with an EMS review used to evaluate the effectiveness of its operation in order to adjust the energy plan/strategy accordingly.

5. Energy Management Systems for Harbours

The benefits of EMS integration in many applications (see Table 2) are undisputed; however, it is necessary to have an organised implementation plan to tackle any challenges that could hinder its optimum operation. As was documented in the previous sections, the

adoption of EMSs by port authorities is gaining ground during recent years in the efforts for carbon intensity reduction and cost minimisation. The maritime sector is responsible for more than 15% of SO_x and NO_x emissions, while maritime emissions exceed 70% of ports' GHGs [72,73]. Therefore, it is apparent that the transition of ports toward sustainability could be deemed of high importance. On top of that, energy consumption in the European port sector has been ranked seventh among the top ten environmental priorities, while, during the last few years, energy consumption lies among the three most important issues, following air quality and climate change [74].

An energy management plan to cope with the variability of the energy demand, increase efficiency, and facilitate the transition to RESs has been developed by the port of Los Angeles, USA. A study revealed that the port's electricity consumption in 2012 was between 200 and 250 TWh at a cost of approximately USD 30 million. The initial assessment based on metering points managed by individual tenants indicated that the highest consumption occurred in containers and bulk terminals. The forecasted energy demand of the port was estimated to exceed 400 TWh by 2020, suggesting the necessity of an EMS able to handle this potential increase. The forecasted increase was based on the one hand on the electrification of the cargo handling equipment and on the other on the potential increase of everyday activities. To this end, the port authorities quoted an energy management strategy consisting of four steps, including organisational foundation, collaboration and outreach, energy master plan studies, and implementing actions. The two first steps were planned to be taken in the short term (1 year), with the third to be implemented in the mid-term (1–2 years), and the last one comprising the infrastructure improvements in the long term (2+ years) [75].

Acciaro et al. presented in their research the contribution of port authorities acting as active energy managers. They used the ports of Hamburg and Genoa as case studies, where an energy and environmental plan estimated $\approx 197,000$ t CO₂ abatement by 2020 (benchmarked to 2011) from an integrated package of measures, and identified a project delivering $\approx 20,000$ t CO₂ annual savings via installation of three solar power stations, cold ironing, and wind energy utilisation, with a total CAPEX of EUR 60 million [76].

Boile et al. as mentioned in the previous section, presented the results of the implementation of energy management plans in six Mediterranean ports under the scope of a European project. According to the authors, less than half of European ports in 2016 have implemented an adequate EMS able to assess through an audit scheme the respective EnPIs and secure an optimised operation. Three levels of intervention were identified in response to the auctions to be held during the development of a port EMS. First, those that the port authority is responsible for; second, those that other entities are responsible for in the wider port area; and, third, the ones that the transport and logistics supply chain is responsible for. The proposed structure of a port's energy management implementation comprises seven steps, as stated below [53]:

- Energy management vision, including aims and objectives.
- Energy policy, regulations, and standards to be taken into consideration.
- Documentation of main energy consumption data.
- Energy demand, potential improvements to secure reliability.
- Selection criteria for improving energy efficiency.
- Identification of the measures to be adopted.
- Timeline/responsibilities of the EMS implementation.

A major technology that is crucial for the efficient implementation of an EMS in harbour applications is the monitoring system that allows communication between the different types of equipment within the port's area. Yang et al. described the importance of contemporary Information Technology (IT) communicating systems for the efficient and

robust operation of a port's EMS. According to the authors, the adoption and utilisation of an IoT approach comprising smart sensors, actuators, wireless equipment, and data centres is essential for the transition of a conventional port to a smart and sustainable one focusing on faster and more efficient services. The layout of such a system is proposed to include four main sections accounting for the main equipment, such as the port's main elements (e.g., cranes, cargo handling, vehicles), the sensing system (e.g., sensors, cameras, inertial navigation modules, Lidar), the communication protocols (e.g., 4G, 5G, GPS, internet), and the control area [77].

A study by Van Duin et al. proposed a six-step benchmarking method to assess energy consumption at port container terminals, while modelling yard-lighting energy at the same area. The authors validated the model at the port terminal of Delta/Rotterdam and estimated a baseline lighting demand of about 6.7 GWh yr⁻¹, quantifying CO₂ at ≈2479 t yr⁻¹ for lighting alone. An HPS→LED lamp retrofit (1.0 kW vs. 0.15 kW per bulb) was taken into consideration and calculated EUR 302 k yr⁻¹ savings, ≈1058 t CO₂ yr⁻¹ avoided, and a ≈2.5-year payback. The authors concluded that the retrofit would cut lighting energy by ≈85%, while also reducing light pollution at the terminal [78].

Ballini and Ölçer examined the application of EMSs in ports based on ISO 50001, and the role of the energy manager to guide the energy team and supervise the development, implementation, and operation of the energy management plan. A case study for the port of Genova was compiled, where the main responsibilities of the energy manager were defined as follows [79]:

- Develop, implement, and maintain the management plan.
- Assess and approve the relevant reports.
- Conduct periodical energy reviews.
- Ensure that targets on EnPIs are met.
- Coordinate energy audits.
- Collect and assess records and management review documents.

Sdoukopoulos et al. reviewed the existing standards, regulations, and measures that can be implemented by port authorities to support energy conservation and sustainability. Among those are the ISO 50001 for EMS, its predecessor EN 16001 (European standard), and the environmental management systems including energy management as one of their main parts. The authors identified four main sectors that are significant to take into account during EMS implementation: the management policy and monitoring systems, along with a responsible team to handle the transition; the prospects of using RES-based technologies in both energy production, energy consumption, and vehicles/equipment utilisation; the sea activities of the vessels and the land activities in regards to vessel loading/unloading; and the intelligent traffic management [80].

Cloquell-Ballester et al. developed an index called 'relevant use of energy' (RUE) and applied it within an ISO 50001 EMS to three terminals in the Port of Valencia. The authors used data from a period between 2008 and 2016 and reported a 29.5% and 40% reduction in energy intensity and energy cost per terminal operation. The outcome of their research proposed that RUE and EMS prioritisation improved energy performance and supported decision making for more efficiency optimisation projects [81].

In this context, Alamoush et al. documented the technical and operational measures that could be adopted by port authorities to improve energy efficiency and consequently reduce GHG emissions. The authors identified two main categories of measures consisting of the portside and ship–port interface. One of the major investigated portside measures is the implementation of EMSs along with the respective technical infrastructure, including RES-based systems, energy storage systems, alternative fuels, smart microgrids, and smart load management [82]. Puig et al. studied the environmental performance of 90 European

ports to identify on the one hand the number of ports that implemented measures to support sustainability, and on the other the type and the target area of the measures. The results showed that 95.7% of the ports sampled had an environmental policy, but 88.9% of the ports operated a monitoring programme, with 80% addressing energy consumption [83].

A study by Durán et al. demonstrated a comparative layout–emissions framework for Chilean container terminals, benchmarking against Valencia, and then estimated the effect of electrification compared to diesel-based processes. For electricity-based scenarios, they estimated a decrease in emissions by 6.9–7.3 t CO₂ depending on terminals under investigation. The results indicated that an EMS-enabled transition from diesel- to electric-powered processes optimised the energy and financial reviewed parameters [84].

Sifakis and Tsoutsos investigated the prospects on near-zero-energy ports by reviewing the available literature and presenting viable solutions that could help port management authorities to identify potential measures to be used for increasing energy efficiency. According to the authors, the implementation of EMSs and the respective measures to cope with the ports' energy-intensive processes requires a conceptual framework that consists of different maturity levels including economic, technological, and managerial experience. Additionally, the results presented showed a priority agenda of the potential measures to be adopted during the implementation of an EMS [85]. In this regard, Sifakis et al. (2021) examined the optimum operation of a hybrid RES in a grid-connected port to secure the lowest possible levelised cost of energy and environmental footprint. The authors proposed a smart EMS operating under 17 scenarios concerning different combinations of mature RES, energy storage systems, and energy management tactics. The case study involved a Greek port with a daily mean energy demand of 7.3 MWh and an electricity bill-based cost of EUR 0.15/kWh. The optimisation methodology used photovoltaic (PV) and wind turbines (WT), with energy-storage-coupled systems based on different battery technologies (i.e., lead acid, lithium-ion, and vanadium redox flow). The results showed that, for a 24 h autonomous system, the levelised cost of electricity was reduced to EUR 0.13/kWh, with the optimal scenario being 10 h of autonomous operation, where the cost of electricity was reduced to EUR 0.08/kWh (scenario of PV 626 kW, wind turbine 100 kW) with a zero carbon footprint. On top of that, the authors argued that peak saving strategy was more efficient in lowering energy bills, while the effect on energy conservation and emissions reduction caused by the implementation of RES and energy storage technologies was highly dependent on the case studied [86].

In the same context, Skok et al. proposed the utilisation of an EMS into harbour's smart grid. The smart grid consisting of conventional power generation (e.g., diesel-fired), RESs, and advanced technologies like fuel cells and power-to-hydrogen systems requires an efficient strategy to overcome problems occurring in managing different energy systems. The solution, according to the authors, of an IT-based EMS Integrator provided energy cost reduction and increased energy supply reliability, while enabling control of its own consumption [87]. Yang analysed the technical and environmental aspects of EMSs in smart ports that utilise IoT, AI, cloud computing, and big data analytics. The author assessed the EMS performance under four case studies regarding the ports of Hamburg, Genoa, Jurong, and Shanghai phase IV. The results indicated a 7–8% reduction in energy consumption, and an 11–12% decrease in carbon emissions. The benefits of EMS implementation stemmed from improvements in load dispatch, transport scheduling, and energy storage [88]. Sadiq et al. surveyed AI/ML studies in maritime power systems with emphasis on smart-port EMS use cases. The authors applied bibliographic/keyword co-occurrence mapping to recent IEEE outputs to classify themes (forecasting, diagnostics, optimisation) and advanced methods (deep learning, reinforcement learning) [89]. Research published by Deng and Han proposed a Fuzzy logic EMS for a green-port multi-energy microgrid integrating renewables

and storage in order to cope with issues arising from the large and frequent fluctuations of power generation and loads. The simulation results showed that the proposed EMS extends the lifespan of the energy storage systems through a maintained state of charge within 34–78% of full capacity [90]. Similarly, Cholidis et al. evaluated hybrid RES portfolios and EMS-optimised storage scheduling to raise port energy autonomy. The optimisation resulted in self-sufficiency, while minimising cost and tested sensitivity to price structure and OPS utilisation. Scenario results indicated $\approx 54\%$ autonomy and $\approx 53\%$ reductions in grid imports [91].

A cost minimisation objective concerning the existing installations of the Norwegian port of Borg was investigated by Massana et al. The research focused on the optimisation of the microgrid operation, consisting of PV production and different loads such as cranes, EV charging, and electrical storage. Real data during the harbour's operations were gathered, and the results showed a 17.2% reduction in operating costs when the optimal scheduler was a part of the EMS [92]. Kelmalis et al. investigated the prospects of 'cold ironing' as a solution to reduce emissions and conventional fuel dependence in the Island of Lesbos. To support the proposed solution, the authors examined the potential of integrating a 20 MW wind farm to meet the electricity demand of the main island's port. The results showed that the optimised allocation of the wind farm's production could cover the energy demand of the harbour equalling to 6.2 GWh/month, while approximately 15 GWh per year could be used either for energy storage or for exporting to the grid, supporting more decreases in CO₂ emissions [93].

In the same context, Vásquez et al. studied how onshore power supply (OPS) to vessels in the port of Sines, Portugal, would reduce the GHG emissions by vessels' manoeuvres. The investigated data spanned the period between 2018 and 2022, and four ports' terminals were considered. The results presented a reduction in energy/emissions baselines under the onshore power supply of vessels, with a sensitivity to grid carbon intensity and RES penetration of 10–25%. However, it is essential to develop appropriate infrastructure, tariff design, and EMS-supported scheduling [94]. Sfair et al., on the other hand, appraised the use of OPS for the port of Santos (Brazil), which has been demonstrated to reduce CO₂, CH₄, and NO_x emissions by 1%, 28%, and 100%, respectively. An EMS-operated measure is essential, though, for shore-connection scheduling, tariff response, and utilisation tracking in order to secure environmental benefits [95]. Likewise, Karimi et al. targeted OPS electrification at busy terminals and proposed a universal shore-to-ship charging architecture serving multiple berths under a supervisory EMS. They specified the power-electronic topology, multi-berth load-sharing, and power-quality controls, and validated performance through case-study simulations [96].

Mohsendokht et al. presented a critical review of seaport resilience between 2000 and 2024 using Web of Science: 1131 records identified; 967 excluded at title/abstract; 164 retained (14%); full-text screening left 128 studies plus 14 added via reference tracking. The research consisted of methods/metrics and disruptive scenarios and identified limited engineering-oriented indicators. The outcome of their study identified requirements on harmonised metrics and more implementation-focused studies (including network-level analyses) to support decision-grade appraisal [97].

Entire-systems policy review mapping, including international/EU/national instruments against local energy system constraints, was investigated by Damman et al. The methodology followed was based on interviews and workshops with four Norwegian ports, plus a systematic reading of policy documents suggesting the major role of adopted policies in harbours' sustainable energy transition [98].

6. Analysis and Discussion on Findings

The empirical relevance of EMS application was enhanced by a meta-analysis summary to synthesise the studies' findings across technologies, sectors, and regions, as the majority of the literature is case-specific or localised. The meta-analysis supported quantitative generalisation by providing statistics on energy savings, cost-benefit ratios, and emission reductions. The summary of the meta-analysis is documented in Table 4.

Table 4. Meta-analysis results summary.

Author	Year	Application	Technology	Benefit Type	Quant Result
Berlad et al. [27]	1976	Residential	Thermal Storage	Cost Reduction	50% reduction in operating costs
Gellings, C. W. [31]	1985	Utility Demand-Side	EMS with DSM integration	Efficiency and Cost Reduction	Up to 20% demand reduction potential identified
Gruber & Brand [36]	1991	SMEs	EMS	Energy Conservation	40–60% energy conservation
St. Clair [35]	1992	Utilities	EMS in Industries and SMEs	Energy Savings	25–35% annual energy savings
Zhang et al. [49]	2013	Residential EMS	EMS + Load Shifting	Cost Reduction	11–23% cost savings; <60% peak demand savings
Acciaro et al. [76]	2014	Port governance/authority	EMS roles and planning	Emissions Reduction (plan-level)	Plan target $\approx 197,000$ tCO ₂ by 2020; ≈ 100 tCO ₂ via 3 PV stations (CAPEX EUR 60 million)
Boile et al. [53]	2016	Port terminal	SCADA + EMS	Energy and Emissions Reduction	250 MWh/year saved; 125 tCO ₂ avoided
Van Duin et al. [78]	2017	Container terminal (Delta, Rotterdam)—yard lighting	LED retrofit within EMS/benchmarking (HPS→LED)	Energy and Cost and Emissions	$\approx 85\%$ ↓ lighting energy; $\approx$ EUR 302 k/yr saved; ≈ 1058 tCO ₂ /yr avoided; ROI ≈ 2.5 yr
Cloquell-Ballester et al. [81]	2020	Container terminals (Port of Valencia)	ISO 50001 EMS + RUE index	Energy Intensity and Cost	29.5% ↓ energy intensity (2008–2016); 40% ↓ energy cost
Hu et al. [57]	2020	EV energy storage	Fuzzy logic EMS	Battery and Range Optimization	6.2% ↑ battery life; 11.1% ↑ range
Sifakis et al. [86]	2021	Port microgrid	EMS + PV/WT + battery storage	Cost and Emissions Reduction	LCOE ↓ to EUR 0.13–0.08/kWh; up to 24 h autonomy

Table 4. *Cont.*

Author	Year	Application	Technology	Benefit Type	Quant Result
Roslan et al. [44]	2021	Perlis, Malaysia	Optimized EMS controller	Cost and Emissions Reduction	62.5% ↓ in cost; 61.98% ↓ in CO ₂
Durán et al. [84]	2022	Chilean terminals (Valparaíso, San Antonio, Arica)	Layout optimisation + electrification (EMS-operable)	Emissions Reduction	Up to 7257 tCO ₂ /yr (San Antonio), 6788 tCO ₂ /yr (Valparaíso), 1282 tCO ₂ /yr (Arica)
Massana et al. [92]	2024	Borg Port (Norway)	PV + EMS + Scheduler	Cost Reduction	17.2% ↓ in operational costs
Yang [88]	2025	Smart Ports (Hamburg, Genoa, Jurong, Shanghai)	IoT + AI + Cloud EMS	Efficiency and Emissions	7–8% ↓ in energy; 11–12% ↓ CO ₂
Vásquez et al. [94]	2025	Port of Sines (Portugal)	Onshore Power Supply + EMS scheduling	Emissions Reduction	Reductions contingent on grid EF and 10–25% RES scenarios
Sfair et al. [95]	2025	Port of Santos (Brazil)	Onshore Power Supply	Emissions Reduction	↓ 1% CO ₂ , 28% CH ₄ , 100% NO _x vs. auxiliary engines
Cholidis et al. [91]	2025	Port autonomy	EMS-optimised storage scheduling	Autonomy/Grid imports	≈54% autonomy; ≈53% ↓ grid imports (scenario)
Romero et al. [42]	2025	Microgrid EMS	Grid EMS	Decreased energy Loss	Decreased energy loss—28% efficiency improvement

To further analyse the findings presented in Table 4 concerning harbour applications, harbour-specific energy savings and cost-effectiveness were synthesised in the following meta-analytic tables (Tables 5 and 6). In Table 5, a harbour-only meta-analytic summary was compiled to enhance cross-study comparison. Reported energy effects were expressed on a common scale (lnRR) for percentage outcomes, while absolute energy saving values were retained as non-pooled entries. It is apparent that direct consumption effects from smart-port EMS deployments were consistent across applications (e.g., 7–8% reduction in multi-port cases), while, when the EMS coordinated energy storage coupled with RESs, grid-electricity imports were approximately reduced by half, suggesting that autonomy-oriented portfolios could materially alter the port’s external energy dependence.

Table 5. Meta-analysis on energy savings in harbours.

Study (Year)	Boile et al. [53]	Van Duin et al. [78]	Cloquell-Ballester et al. [81]	Yang [88]	Cholidis et al. [91]
Port	Port of Koper (terminal)	Port of Rotterdam (terminal)	Port of Valencia (3 terminals)	Ports (Hamburg, Genoa, Jurong, Shanghai)	Generic port (autonomy scenarios)
EMS/Measure	SCADA + EMS (monitoring/optimisation)	LED retrofit within EMS/benchmarking (HPS→LED)	ISO 50001 EMS+ RUE index	IoT + AI + Cloud EMS	Hybrid RES + EMS-optimised storage scheduling
Application	Terminal/site	Yard lighting	Energy intensity	Port-level operations	Site-level (grid imports proxy)
Metric	Absolute ΔE	%Δ lighting energy	%Δ energy intensity	%Δ energy	%Δ energy (grid imports)
Unit	MWh/yr	%	%	%	%
Value	−250	−85	−29.5	−7.5	−53
InRR	-	−1.897	−0.35	−0.078	−0.755
Notes	EUR 65 k EMS; EUR 19.5 k/yr savings; 125 tCO ₂ /yr avoided.	Lighting-only; baseline ≈ 6.7 GWh/yr; ROI ≈ 2.5 yr.	Energy cost ↓ 40%.	Reported 7–8% energy reduction (midpoint used for InRR).	≈54% autonomy; ≈53% ↓ grid imports (proxy for site energy).

Table 6. Cost-effectiveness of EMS applications in harbours.

Study (Year)	Boile et al. [53]	Van Duin et al. [78]	Sifakis et al. [86]	Acciaro et al. [76]
Port	Port of Koper (terminal)	Port of Rotterdam (terminal)	Greek port (microgrid)	Port of Genoa (plan-level action)
EMS/Measure	SCADA + EMS (monitoring/optimisation)	LED retrofit within EMS/benchmarking (HPS→LED)	Hybrid RES + EMS	Energy and environmental plan incl. PV
CAPEX (M€)	0.065	0.77	3.4–5.0	60
Annual savings (EUR)	19,500	302,000	≈400,000	-
Simple payback (yr)	3.3	2.5	6–12	-
LCOE (€/kWh)	-	-	0.08–0.13	-
Energy outcome	≈250 MWh/yr saved	≈85% ↓ lighting energy	24 h autonomy scenario feasible	-
Emissions outcome	≈125 tCO ₂ /yr avoided	≈1058 tCO ₂ /yr avoided	≈6000 tCO ₂ /yr avoided	≈20,000 tCO ₂ /yr; ≈197,000 tCO ₂ plan target to 2020
Annual benefit per tCO ₂ (€/t·yr)	156	285.4	66.6	304.5

The log response ratio indicated that lighting retrofit gives a substantial decrease in energy consumption, followed by the utilisation of RES–EMS storage implementation. However, one should take into account that the lowest score achieved by the use of IoT, AI, and cloud technologies still poses a great effect considering that the energy conservation concerns the entire port operations.

Table 6 shows that EMS-type measures in harbours are economically attractive whenever direct monetary savings are reported. Payback periods are into the acceptable range, especially when low investments are adopted. Regarding the annual cost benefit per emis-

sions avoided it seems that low targeted investments (e.g., EMS optimisation, lighting retrofit) are in favour. However, the largest benefit arises when onshore power supply (cold ironing) is included in the plan (e.g., Acciaro et al.), where vessel emissions are avoided in parallel to the emissions from the electricity consumption.

6.1. Recommended Techniques for Energy Conservation in Ports

Based on Table 3, the third maturity level of EMS implementation includes the implementation of the necessary equipment that would facilitate the transition of a business toward an environmentally friendlier profile. Likewise, port operations, as described above, present significant energy intensive processes and measures for mitigating the impacts associated with those that are investigated in the international literature. By taking into account the quoted literature and the data from the meta-analysis documented in Tables 4–6, Table 7 documents the most significant measures that are adopted during the implementation of a port's EMS presenting additionally the associated benefits or challenges.

Table 7. Technologies/systems/policies implemented during EMS development in ports.

Technologies/Measures	Description	Benefits	Challenges	Sources
Peak shaving/load shifting	Shift in the energy demand to off-peak periods	Lower electricity costs, low investment	Efficient allocation of energy to meet the demands	[78,99]
RES based lighting and HVAC systems	LED lights, smart sensors, and electrification of HVAC systems (e.g., heat pumps)	Lower energy/electricity costs, high efficiencies	High capital cost, proper sizing to meet the demands	[78,100]
On-shore power, cold ironing	Use of shore electricity by berthed vessels instead of using diesel generators	Reduction in the port's carbon footprint, income for the port authorities	High capital cost, efficient scheduling to provide competitive electricity prices	[94,95,101,102]
Electrification/hybridisation of container/cargo handling equipment and vehicles	Replacement of conventional cranes and cargo handling equipment with electric-based ones	Reduction in the port's carbon footprint, high efficiencies, lower energy costs, less noise, automation	High capital cost, higher electricity costs—proper management of electricity use (based on electricity contracts, RES implementation)	[54,73,84]
RES-based electricity generation	Installation of PVs on rooftops, WTs to unallocated fields or offshore, wave energy	Lower electricity costs, reduction in port's carbon footprint, income for port authorities (selling electricity back to the grid)	High capital cost, proper management of electricity use to meet the demands, stochastic operation, maturity matters (wave energy)	[55,85]
Energy storage systems	Topologies able to store electricity (e.g., batteries, flywheels, ultra-capacitors, hydrogen)	Electricity reliability in case of shortages, coupled with RES facilitates the above recommended measures implementation	High capital cost, maturity matters, proper management of the energy flows allocation	[54,73]
Oil waste utilisation	Use of waste oils in thermal energy applications	Lower costs in space conditioning, support of circular economy	High capital costs	[103]
Alternative fuels	Use of alternative fuels (e.g., LNG, methanol, biofuels, biomass, hydrogen) for combined heat and power (CHP), vehicles and other motorised equipment	Reduction in the port's carbon footprint, low-cost fuels, high efficiency for heating/power applications	High capital costs, proper management of fuel availability	[54,85]
Real-time energy monitoring system	Use of IoT, smart sensors, actuators, wireless devices and data centres to measure, collect and store energy consumption data	Provides all necessary data on real time to assess the operation of equipment, higher operation efficiency, automated equipment control	Relatively high capital cost, personnel training is required, equipment sensitivity, data quality, cybersecurity risks	[54,77,88]

The assessment of the above documented technologies in terms of integration in harbour activities has been processed through a WSM ranking analysis. Equation (1) was used by taking into account four criteria including energy conservation, the environmental benefit of benefits, the implementation cost, and the complexity of challenges. The weights of each criterion have been assumed to be equal to 0.25 (25%), and the value fraction was qualitatively determined based on the given data from Table 7 within a range of 1–5, with 1 indicating minimum and 5 the maximum effect of the criteria, respectively. The scores 1–5 were linearly rescaled by division by 5 in order to make weights interpretable, prevent any criterion from exerting implicit extra influence due to scale differences when aggregating benefits and challenges, and finally enable sensitivity analysis. To clarify the outcome of the WSM by aggregating benefits and challenges, a comparable indicator was deployed by computing a ‘Net’ score (benefits minus challenges score). Positive values indicate that weighted benefits outweigh weighted challenges; negative values indicate the opposite. To rank the measures in the case of equal ‘Net’ scores, a ‘Tiekey’ formula was implemented. This formula calculates a number for each measure, only to break ties when two (or more) measures have the same displayed ‘Net’ score. It preserves a priority order among secondary criteria without changing any reported values. The rule used comprised the following statements:

- Highest Net wins.
- If Net ties, prefer higher benefit.
- If still tied, prefer lower cost.
- If still tied, prefer lower complexity.

The relevant data are presented in Table 8.

Table 8. WSM baseline data and results on EMS measures/policies for harbours.

Criteria	Energy Conservation	Environmental Benefit	Implementation Cost	Complexity	Benefits Score	Challenges Score	Net Score	Tiekey	Rank
Weights of criteria (W_i)	0.25	0.25	0.25	0.25					
Technologies/measures	Value function (X_i)								
Peak shaving/load shifting	2	2	1	2	1	0.75	0.25	0.250000760	2
RES based lighting and HVAC systems	3	3	5	2	1.5	1.75	−0.25	−0.249999540	9
On-shore power, cold ironing	4	4	4	3	2.0	1.75	0.25	0.250001140	1
Electrification/hybridisation of container/cargo handling equipment and vehicles	3	4	5	3	1.75	2.0	−0.25	−0.249999310	6
RES-based electricity generation	4	4	5	4	2.0	2.25	−0.25	−0.249999080	5
Energy storage systems	3	3	4	3	1.5	1.75	−0.25	−0.249999360	8
Oil waste utilisation	2	5	5	3	1.75	2.0	−0.25	−0.249999310	6
Alternative fuels	3	5	5	3	2.0	2.0	0	0.000000940	3
Real-time energy monitoring system	3	3	3	3	1.5	1.5	0	0.000000840	4

The following figure (Figure 9) indicates the results of the WSM analysis. The results are presented in a scatter diagram where, in conjunction with Table 8, policy-decision managers may exclude conclusions in regards to the most beneficial technique to be followed.

According to Table 3, the implementation of EMS measures/policies in harbours is highly dependent on the maturity level of the EMS implementation. Hence, based on Table 8 and Figure 9, it is apparent that, at the initial maturity levels of EMS integration in harbour activities, the most appealing policy seems to be the ‘Peak Shaving/Load Shifting’ measure, which scores the lowest in the challenging axis, and in a medium score in the benefits, while ranking 2 in the WSM. Likewise, ‘Real-time Monitoring’ presents a good potential of a measure to be adopted, as it provides benefits without a prohibitive cost,

ranking 4 in the WSM. On the other hand, in more advanced maturity levels, ‘On-shore Power (Cold Ironing)’ seems also to be a beneficial case as, although it scores relatively high in the challenges, it presents a high benefit score that outcomes the challenges arising from its adoption and ranks 1 as the best measure to adopt. Ranks 5 to 9 present negative ‘Net’ scores, indicating that challenges outweigh benefits and could be deemed as solutions to be adopted in a later stage of port EMS strategies.

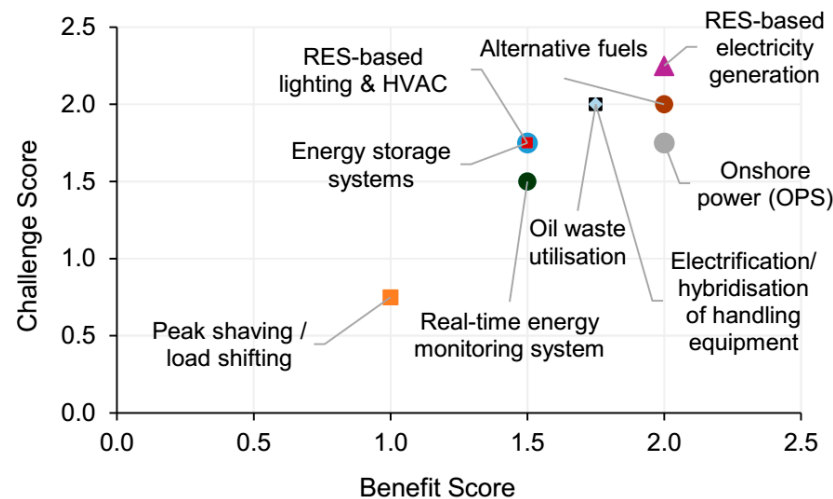


Figure 9. Results of the baseline WSM analysis for EMS policies in harbours.

To enhance the robustness of the WSM rankings, a Monte Carlo sensitivity analysis was performed in ‘Python’ to assess all the plausible combinations. First, preference uncertainty was examined by varying the criterion weights across four policy-relevant scenarios (balanced; benefit-focused; cost-focused; energy-tilted), while keeping the underlying 1–5 scores fixed. Three weight scenarios were investigated, including S1 Benefit-focused (emphasised Energy and Environment), S2 Cost-focused (emphasised Cost and Complexity), and S3 Energy-tilt (favoured Energy while reducing weights from the other). The weights for each scenario are presented in Table 9.

Table 9. Weights of criteria under different sensitivity analysis scenarios.

Scenario	Weights of Criteria (W_i)			
	Energy Conservation	Environmental Benefit	Implementation Cost	Complexity
S1 Benefit-focused	0.35	0.35	0.15	0.15
S2 Cost-focused	0.15	0.15	0.40	0.30
S3 Energy-tilt	0.50	0.20	0.20	0.10

The next figures (Figures 10–12) indicate the WSM results of the weight criteria sensitivity analysis.

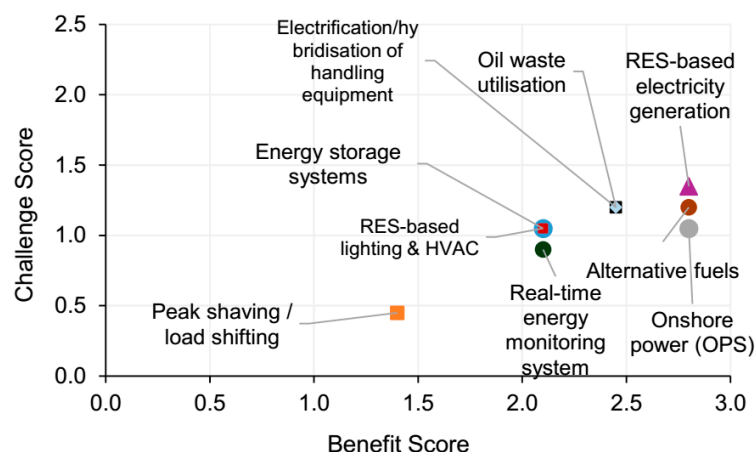


Figure 10. Results of the S1 WSM analysis for EMS policies in harbours.

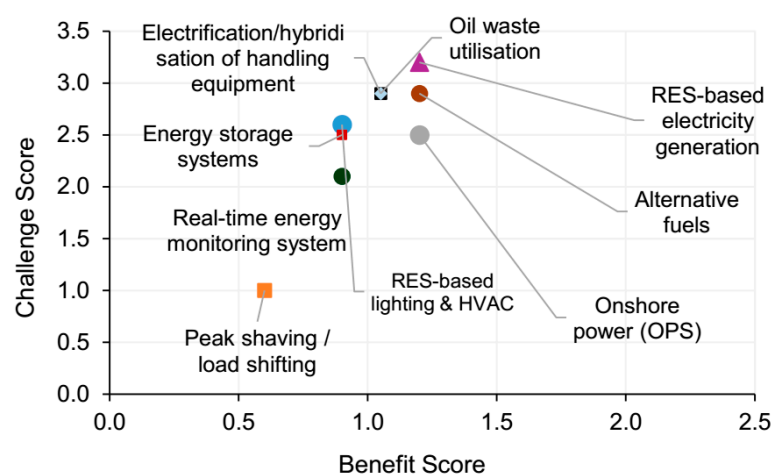


Figure 11. Results of the S2 WSM analysis for EMS policies in harbours.

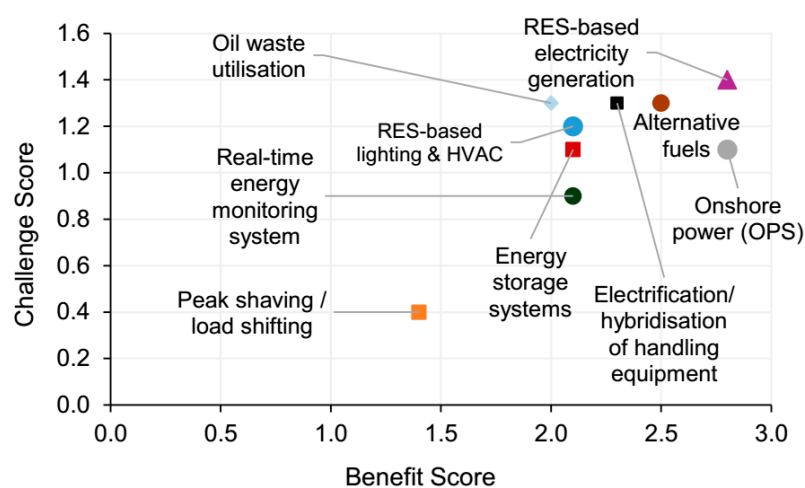


Figure 12. Results of the S3 WSM analysis for EMS policies in harbours.

The results depicted in Figure 13 are summarised, showing how each EMS measure is ranked among the baseline and the three weight criteria analysis.

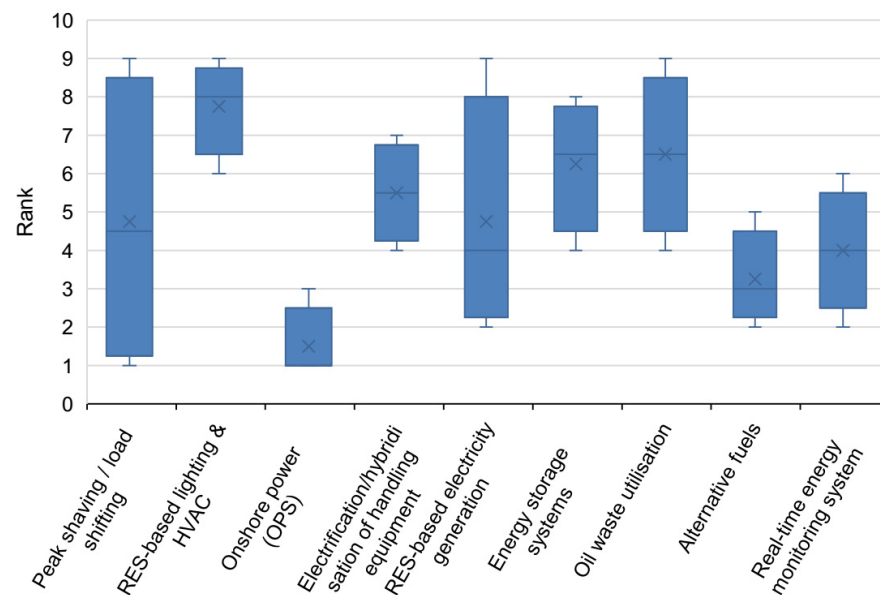


Figure 13. Rank scores and median values during the weight of criteria sensitivity analysis.

It is obvious that the measure that seems to score best among all the cases is the ‘OPS (Cold Ironing)’, followed by the ‘Alternative fuels’ and the ‘Real-time energy monitoring system’. The measure that seemed to be the best in the baseline scenario ranked 1 in scenario 2 (Cost-focused), as it is the most cost-effective, but presented a higher median value compared to others.

Second, score uncertainty was probed with a one-at-a-time procedure in which, for each measure and criterion (Energy, Environmental, Cost, Complexity), the original score was perturbed by integer steps $\Delta \in \{-3, -2, -1, +1, +2, +3\}$, with values clamped to 1–5 to avoid inadmissible scores. Third, a Monte Carlo experiment drew integer scores uniformly within ± 3 of each baseline score (also clamped to 1–5) over a large number of trials, recomputing Benefits, Challenges, ‘Net’ score, and ranks. Reported outcomes included rank, top-1 and top-3 frequencies, mean Net scores, and rank distributions. Table 10 presents the rank frequency of each measure during the Monte Carlo sensitivity analysis (green: for low frequency, orange: for average frequency, red: for high frequency). Figure 14 shows the top-1 to 3 measure frequencies to identify the best measures to be initially adopted during harbours EMS implementation.

Table 10. Frequencies of EMS measure ranks during the Monte Carlo analysis.

Measure	Rank1	Rank2	Rank3	Rank4	Rank5	Rank6	Rank7	Rank8	Rank9
Peak shaving/load shifting	0.152	0.123	0.111	0.134	0.117	0.115	0.096	0.082	0.070
RES-based lighting and HVAC	0.095	0.091	0.099	0.116	0.109	0.111	0.120	0.118	0.141
Onshore power (OPS)	0.117	0.126	0.108	0.104	0.104	0.104	0.087	0.131	0.119
Electrification/hybridisation of handling equipment	0.109	0.101	0.112	0.080	0.088	0.125	0.141	0.116	0.128
RES-based electricity generation	0.083	0.092	0.105	0.108	0.106	0.101	0.121	0.139	0.145
Energy storage systems	0.130	0.106	0.091	0.102	0.124	0.097	0.126	0.116	0.108
Oil waste utilisation	0.114	0.125	0.12	0.123	0.119	0.119	0.100	0.083	0.097
Alternative fuels	0.092	0.128	0.127	0.122	0.125	0.111	0.106	0.112	0.077
Real-time energy monitoring system	0.120	0.120	0.125	0.119	0.100	0.117	0.104	0.096	0.099

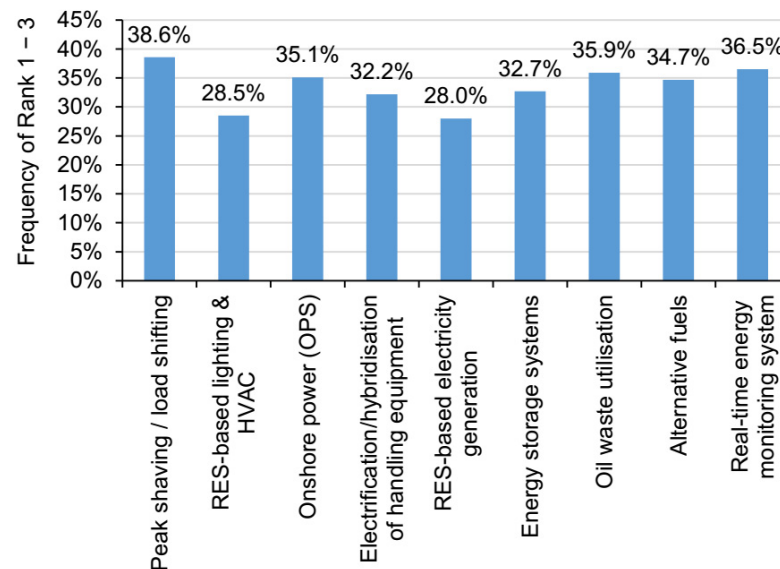


Figure 14. Frequencies of EMS measures top three ranks during the Monte Carlo analysis.

Based on Table 10 and Figure 14, the ‘Peak shaving/load shifting’ EMS measure ranked in the first three positions most of the times under the baseline weight of criteria and a Monte Carlo sensitivity analysis on the Benefits and Challenges scores. Second in line scored the ‘Real-time monitoring system’, followed by ‘Oil waste utilisation’ and ‘OPS (Cold ironing)’. The analysis therefore shows that the most effective EMS measure in all aspects/criteria (energy, environment, cost, complexity) seems to be the ‘Peak shaving/load shifting’. However, it should not be ignored that ‘OPS (Cold ironing)’ could outweigh it in the case of different weights, and particularly when energy and environmental aspects are favoured by stakeholders.

6.2. Discussion on the Findings

Across harbour deployments, EMS adoption consistently reduced operational energy at site level; multi-port studies reported $\approx 7\text{--}8\%$ reductions when EMS optimised dispatch and scheduling, while targeted yard-lighting retrofits within EMS benchmarking achieved $\approx 85\%$ cuts in lighting loads with short paybacks. These effects were corroborated by terminal cases reporting absolute savings ($\approx 250 \text{ MWh yr}^{-1}$) and associated emission reductions. When EMS coordinated storage with RES, the outcome shifted from incremental savings to structural reliance changes with scenarios indicating up to 54% energy autonomy. Onshore power supply, on the other hand, emerged as a high-impact environmental measure when EMS scheduling and tariff design were present.

Rankings were stable to plausible changes in criteria weights and score uncertainty. Monte Carlo experiments showed ‘Peak-shaving/load-shifting’ and ‘Real-time monitoring’ more frequently in the top-three, with OPS competitive and capable of dominating when benefit-oriented preferences were emphasised.

Taken together, the evidence indicated the following:

- Consistent direct savings from EMS and EMS-enabled retrofits.
- Step-change autonomy potential when EMS controls RES + storage.
- EMS scheduling and monitoring techniques consist of attractive financial investments, with OPS yielding the largest absolute abatement under supportive energy and environmental policy.

7. Conclusions

This study presented a literature review on EMSs focusing on their application in harbour environments. The review revealed an increasing academic interest in EMSs, specifically after the development of the electronic technology unlocking the full potential of EMS implementation. The trend of growing global relevance of EMS research since the 2000s is connected with the energy conservation and environmental awareness dominating academic interests during the last 20 years. In the context of the first RQ, the literature review indicated the main applications of EMSs, which initially focused on the building sector and subsequently on industries and businesses aiming to mainly reduce operational expenditures. The shifting to a sustainable energy approach in most contemporary sectors, in conjunction with new emerging technologies in electricity generation and transportation, created a new field of EMS application, including RES integration, electric vehicles, TSOs–DSOs energy management, and harbour processes. What is of high importance during the implementation of EMSs is the maturity level of integrating systems into an application. Specifically, for EMSs, five levels have been identified underscoring the technological and policy dimensions that define EMS adoption.

The second part, aiming to give insights concerning the second RQ, followed a comprehensive literature review on EMS application on harbour activities. Through a meta-analysis of the reviewed sources across applications identified from RQ 1, benefits of EMS integration have been quoted, providing indicative outcomes in energy savings and emissions reduction. The benefits and challenges of EMS integration measures/policies in harbour processes were therefore documented and subsequently assessed via a WSM and sensitivity analyses on weight criteria and scores to highlight the most applicable option depending on the maturity level of EMS implementation in port activities.

To enhance the novelty of this research, a comparative table (Table 11) between this research and other reviews on the specific area of interest is presented.

Table 11. Comparison of areas of interest between previous reviews and current review.

Study	Scope/Sector(s)	Port Focus	Methods	Period Covered	Quant Synthesis	Standards/Policy	What This Work Adds
Lee & Cheng [7]	EMS energy savings (general)	No	Narrative review	1980s–2010s	No	–	Added port-specific meta-summary and WSM.
Zia et al. [6]	Microgrid EMS methods	No	Critical review	– 2018	No	–	Harbour operations with quantitative effects.
Iris & Lam [54,55]	Port operations and energy	Yes	Review + optimisation framing	– 2021	No	Smart-grid/RES	Contributed pooled metrics and cost-effectiveness.
Sdoukopoulos et al. [80]	EU port practices	Partial	Review	– 2019	No	ISO/EMS practice	Added PRISMA statement + ranking robustness.
Alamouh et al. [82]	Port GHG measures	Yes	Review	– 2020	No	–	Identification of OPS/lighting/RES impacts across studies.
Sifakis & Tsoutsos [85]	Near-zero energy ports	Yes	Review + scenarios	– 2021	Partial (scenarios)	–	Meta-summarise realised cases and costs.
Sadiq et al. [89]	AI/ML in maritime power systems	Partial	Bibliographic/bibliometric analysis	– 2025	Counts and networks (no meta-analysis)	–	Added harbour-specific pooling, cost-per-tCO ₂ , and WSM ranking.
Mohsendokht et al. [97]	Seaport resilience (ports, networks)	Yes	Critical review	2000–2024	No	Resilience frameworks and policy	Focused on EMS/energy outcomes and sequencing; connected resilience insights to EMS maturity roadmap.
This review	EMS evolution + harbours	Yes	PRISMA + meta-summary + WSM + sensitivity	1973–2025	Yes	ISO 50001/PDCA	Empirical pooling, cost per tCO ₂ , ranked measures with robustness.

In conclusion, this research confirms that EMSs comprise a cornerstone in sustainability for most energy intensive sectors. Particularly, harbour processes presenting high energy consumption benefit significantly from the adoption of EMSs through enhanced energy control and reduced emissions. However, it is important to apply policies and measures in line with the maturity level in terms of energy management to get a successful outcome in both financial and sustainable fields.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
CAPEX	Capital expenditure
CHP	Combined heat and power
CMMI	Capability Maturity Model Integration
DSO	Distribution system operator
EMS	Energy management system
EnPIs	Energy performance indicators
Eq	Equation
EV	Electric vehicle
GHG	Greenhouse gas
HPS	High pressure sodium
HVAC	Heating, ventilation and air conditioning
IoT	Internet of Things
ISO	International organisation for standardisation
IT	Information Technology
LCOE	Levelised cost of energy
LED	Light emission diode
ML	Machine learning
NN	Neural network
OPS	Onshore power supply
PDCA	Plan, do, check, act
PEMFC	Proton exchange membrane fuel cell
PV	Photovoltaic
RES	Renewable energy systems
RQ	Research question
ROI	Return of investment
RUE	Relevant use of energy
SCADA	Supervisory control and data acquisition
SMES	Small and medium enterprises
TSO	Transmission system operator
UNIDO	United nations industrial development organisation
WSM	Weighted sum model
WT	Wind turbine

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